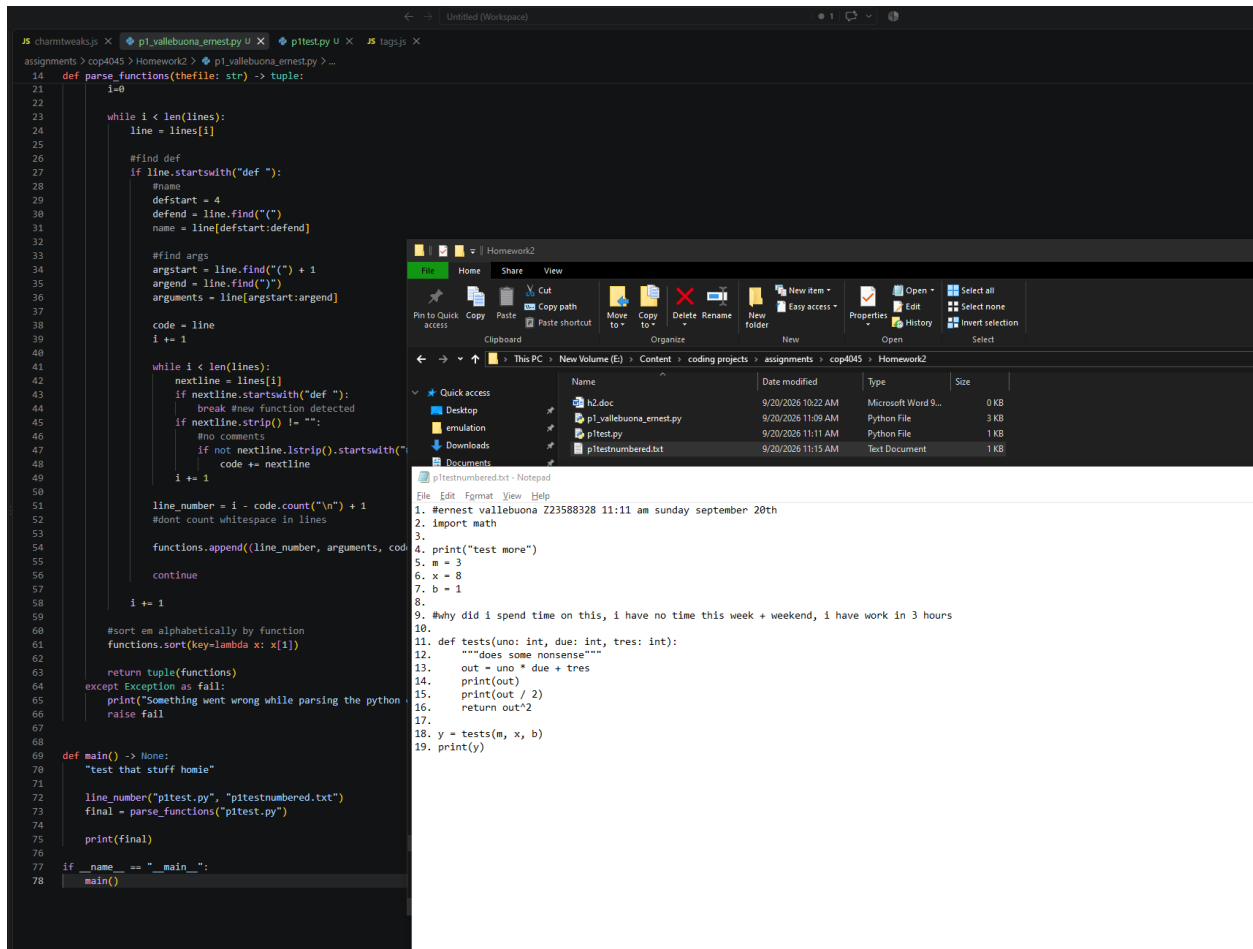


Ernest Vallebuona

Just put comments in the comments on canvas,

Problem 1



def line_number(input_file: str, output_file: str) -> None:

"""read a text file and record each line alongside a line number"""

try:

with open(input_file, "r") as fin:

with open(output_file, "w") as fout:

```
    num = 1

    for line in fin:

        fout.write(str(num) + ". " + line)

        num += 1

except Exception as fail:

    print("Something went wrong while numbering the lines: ")

    raise fail
```

def parse_functions(thefile: str) -> tuple:

```
    """read a Python file and give information about functions within it"""
```

```
    try:
```

```
        with open(thefile, "r") as file:
```

```
            lines = file.readlines()
```

```
        functions = []
```

```
        i=0
```

```
        while i < len(lines):
```

```
            line = lines[i]
```

```
            #find def
```

```
            if line.startswith("def "):
```

```
                #name
```

```
                defstart = 4
```

```
                defend = line.find("(")
```

```
                name = line[defstart:defend]
```

```
#find args
```

```
argstart = line.find("(") + 1
```

```
argend = line.find(")")
```

```
arguments = line[argstart:argend]
```

```
code = line
```

```
i += 1
```

```
while i < len(lines):
```

```
    nextline = lines[i]
```

```
    if nextline.startswith("def "):
```

```
        break #new function detected
```

```
    if nextline.strip() != "":
```

```
        #no comments
```

```
        if not nextline.lstrip().startswith("#"):
```

```
            code += nextline
```

```
    i += 1
```

```
line_number = i - code.count("\n") + 1
```

```
#dont count whitespace in lines
```

```
functions.append((line_number, arguments, code))
```

```
continue
```

```
i += 1
```

```
#sort em alphabetically by function
```

```
functions.sort(key=lambda x: x[1])
```

```
return tuple(functions)
```

```
except Exception as fail:
```

```
    print("Something went wrong while parsing the python code")
```

```
    raise fail
```

```
def main() -> None:
```

```
    "test that stuff homie"
```

```
line_number("p1test.py", "p1testnumbered.txt")
```

```
final = parse_functions("p1test.py")
```

```
print(final)
```

```
if __name__ == "__main__":
```

```
    main()
```

Problem 2

```
53
54 # f)
55
56 text = "Hello world"
57
58 part_f = {
59     1: c
60     for i, c in enumerate(text)
61     if c.lower() in "aeiou"
62 }
63
64
65
66 print("a)", part_a)
67 print("b)", part_b)
68 print("c)", part_c)
69 print("d)", part_d)
70 print("e)", part_e)
71 print("f)", part_f)
72 print("Ernest vallebyona: 223588328 ")
```

Problems 456 Output Debug Console Terminal Ports

PS E:\Content\coding_projects> & C:\Users\Verme\AppData\Local\Microsoft\WindowsApps\python3.13.exe "e:/Content/coding_projects/assignments/cop4845/homework2/p2_Vallebyona_Ernest.py"

a) [(1, 8, 4, 7), (1, 8, 7, 4), (2, 9, 6, 7), (2, 9, 7, 6), (4, 7, 1, 8), (4, 7, 8, 1), (6, 7, 2, 9), (6, 7, 9, 2), (7, 4, 1, 8), (7, 4, 8, 1), (7, 6, 2, 9), (7, 6, 9, 2), (8, 1, 4, 7), (8, 1, 7, 4), (9, 2, 6, 7), (9, 2, 7, 6)]

b) [('one', 3), ('two', 3), ('ten', 3)]

c) ['Christopher A. Kutcher', 'Elizabeth S. Fey']

d) [('Trams', 'Smart'), ('Elbows', 'Bowels'), ('Tops', 'Stop'), ('Astral', 'Altars')]

e) ('one': 3, 'two': 3, 'three': 5)

f) {'e': 4, 'o': 7, 'a': 1}

Ernest vallebyona: 223588328

PS E:\Content\coding_projects>

a)

part_a = [

(a, b, c, d)

for a in range(1, 11)

for b in range(1, 11)

for c in range(1, 11)

for d in range(1, 11)

if len({a, b, c, d}) == 4

and a**2 + b**2 == c**2 + d**2

]

b)

```
strings = ['One', 'SEVEN', 'three', 'two', 'Ten']
```

```
part_b = [  
    (word.lower(), len(word))  
    for word in strings  
    if len(word) < 5  
]
```

c)

```
names = ['Christopher Ashton Kutcher', 'Elizabeth Stamatina Fey']
```

```
part_c = [  
    f"{first} {middle[0]}. {last}"  
    for first, middle, last in (name.split() for name in names)  
]
```

d)

```
lst1 = ["Spam", "Trams", "Elbows", "Tops", "Astral"]
```

```
lst2 = ["Bowels", "Sample", "Altars", "Stop", "Course", "Smart"]
```

```
part_d = [
```

```
(w1, w2)
for w1 in lst1
for w2 in lst2
if sorted(w1.lower()) == sorted(w2.lower())
]
```

```
# e)
s = ['one', 'two', 'three']
```

```
part_e = {
    word: len(word)
    for word in s
}
```

```
# f)
```

```
text = "Hello world"
```

```
part_f = {
    i: c
    for i, c in enumerate(text)
    if c.lower() in "aeiou"
}
```

```
print("a)", part_a)
```

```
print("b)", part_b)
```

```
print("c)", part_c)
```

```
print("d)", part_d)
```

```
print("e)", part_e)
```

```
print("f)", part_f)
```

```
print("Ernest vallebyona: Z23588328 ")
```

Problem 3

```
chumtweaks.py x p2_vallebuona_ernest.py u x p3_vallebuona_ernest.py u x p4_vallebuona_ernest.py u x tags.py x
assignments > cop4045 > Homework2 > p3_vallebuona_ernest.py > ...
89 def load_network(filename: str) -> dict:
102     sn[username] = (fullname, friends)
103
104     return sn
105
106 except Exception as fail:
107     print("Error. Cannot load: ", fail)
108     raise
109
110
111 def main() -> None:
112     """Test all functions."""
113
114     try:
115         sn = {
116             "alice": ("Alice Smith", ["maria"]),
117             "maria": ("Maria Cortez", ["alice", "joe", "david"]),
118             "joe": ("Joseph Adams", ["maria", "eve"]),
119             "eve": ("Evelyn Cooper", ["joe"]),
120             "david": ("David Benson", ["maria"])
121         }
122
123         print("Original network:")
124         print(sn)
125
126         print("\nTesting add_user:")
127         print(add_user(sn, "john", "John Doe"))
128         print(add_user(sn, "alice", "Alice Jones"))
129
130         print("\nTesting add_friend:")
131         print(add_friend(sn, "john", "alice"))
132         print(add_friend(sn, "john", "bob"))
133
134         print("\nTesting get_friends:")
135         print("Alice, distance 1:", get_friends(sn, "alice", 1))
136         print("Alice, distance 2:", get_friends(sn, "alice", 2))
137
138     except:
139         print("Error in main function.")
140
141 if __name__ == "__main__":
142     main()
```

Problems Output Debug Console Terminal Ports

PS E:\Content\coding projects> & C:\Users\Ernest\AppData\Local\Microsoft\WindowsApps\python3.13.exe "e:\Content\coding projects\assignments\cop4045\Homework2\p3_vallebuona_ernest.py"

Original network:

```
{'alice': ('Alice Smith', ['maria']), 'maria': ('Maria Cortez', ['alice', 'joe', 'david']), 'joe': ('Joseph Adams', ['maria', 'eve']), 'eve': ('Evelyn Cooper', ['joe']), 'david': ('David Benson', ['maria'])}
```

Testing add_user:

```
True
False
```

Testing add_friend:

```
True
False
```

Testing get_friends:

```
Alice, distance 1: ['maria', 'john']
Alice, distance 2: ['maria', 'john', 'joe', 'david']
Alice, distance 3: ['maria', 'john', 'joe', 'david', 'eve']
Unknown user: []
```

Testing save_network:

```
Network saved successfully.
```

Testing load_network:

```
{'alice': ('Alice Smith', ['maria', 'john']), 'maria': ('Maria Cortez', ['alice', 'joe', 'david']), 'joe': ('Joseph Adams', ['maria', 'eve']), 'eve': ('Evelyn Cooper', ['joe']), 'david': ('David Benson', ['maria']), 'john': ('John Doe', ['alice'])}
```

Ernest vallebuona: 223588328

PS E:\Content\coding projects>

import csv

def add_user(sn: dict, username: str, fullname: str) -> bool:

"""Add a new user to the network."""

try:

```
if username in sn:
```

```
    return False
```

```
sn[username] = (fullname, [])
```

```
return True
```

```
except Exception as fail:
```

```
    print("Error. cannot add user: ", fail)
```

```
    raise
```

```
def add_friend(sn: dict, user1: str, user2: str) -> bool:
```

```
    """Add a friendship between two users <3."""
```

```
    try:
```

```
        if user1 not in sn or user2 not in sn:
```

```
            return False
```

```
        if user1 == user2:
```

```
            return False
```

```
        if user2 not in sn[user1][1]:
```

```
            sn[user1][1].append(user2)
```

```
        if user1 not in sn[user2][1]:
```

```
            sn[user2][1].append(user1)
```

```
return True
```

```
except Exception as fail:
```

```
    print("Error. Cannot Friend: ", fail)
```

```
    raise
```

```
def get_friends(sn: dict, user1: str, distance: int) -> list:
```

```
    """Return all friends within a distance."""
```

```
    try:
```

```
        if user1 not in sn or distance <= 0:
```

```
            return []
```

```
        visited = {user1}
```

```
        current = [user1]
```

```
        result = []
```

```
        for i in range(distance):
```

```
            next_level = []
```

```
            for user in current:
```

```
                for friend in sn[user][1]:
```

```
                    if friend not in visited:
```

```
                        visited.add(friend)
```

```
                        result.append(friend)
```

```
                        next_level.append(friend)
```

```
current = next_level
```

```
if len(current) == 0:
```

```
    break
```

```
return result
```

```
except Exception as fail:
```

```
    print("You have no friends, or less likely, an error occurred:", fail)
```

```
    raise
```

```
def save_network(filename: str, sn: dict) -> None:
```

```
    """Save the social network to disk."""
```

```
    try:
```

```
        with open(filename, "w", newline="") as file:
```

```
            writer = csv.writer(file)
```

```
            for username in sn:
```

```
                fullname = sn[username][0]
```

```
                friends = sn[username][1]
```

```
                writer.writerow([username, fullname] + friends)
```

```
    except Exception as fail:
```

```
print("Error. Cannot save: ", fail)
```

```
raise
```

```
def load_network(filename: str) -> dict:
```

```
    """Load a social network from disk."""
```

```
    try:
```

```
        sn = {}
```

```
        with open(filename, "r", newline="") as file:
```

```
            reader = csv.reader(file)
```

```
            for row in reader:
```

```
                username = row[0]
```

```
                fullname = row[1]
```

```
                friends = row[2:]
```

```
                sn[username] = (fullname, friends)
```

```
        return sn
```

```
    except Exception as fail:
```

```
        print("Error. Cannot load: ", fail)
```

```
        raise
```

```
def main() -> None:
```

```
    """Test all functions."""
```

```
    try:
```

```
        sn = {
```

```
            "alice": ("Alice Smith", ["maria"]),
```

```
            "maria": ("Maria Cortez", ["alice", "joe", "david"]),
```

```
            "joe": ("Joseph Adams", ["maria", "eve"]),
```

```
            "eve": ("Evelyn Cooper", ["joe"]),
```

```
            "david": ("David Benson", ["maria"])
```

```
        }
```

```
        print("Original network:")
```

```
        print(sn)
```

```
        print("\nTesting add_user:")
```

```
        print(add_user(sn, "john", "John Doe"))
```

```
        print(add_user(sn, "alice", "Alice Jones"))
```

```
        print("\nTesting add_friend:")
```

```
        print(add_friend(sn, "john", "alice"))
```

```
        print(add_friend(sn, "john", "bob"))
```

```
        print("\nTesting get_friends:")
```

```
        print("Alice, distance 1:", get_friends(sn, "alice", 1))
```

```
        print("Alice, distance 2:", get_friends(sn, "alice", 2))
```

```
print("Alice, distance 3:", get_friends(sn, "alice", 3))
```

```
print("Unknown user:", get_friends(sn, "bob", 2))
```

```
print("\nTesting save_network:")
```

```
save_network("social_network.csv", sn)
```

```
print("Network saved successfully.")
```

```
print("\nTesting load_network:")
```

```
new_sn = load_network("social_network.csv")
```

```
print(new_sn)
```

```
except Exception as fail:
```

```
    print("Something went wrong: ", fail)
```

```
    raise
```

```
print("Ernest vallebyona: Z23588328 ")
```

```
if __name__ == "__main__":
```

```
    main()
```

Problem 4

```
JS charmtweaks.js X p2_vallebuona_ernest.py U X p3_vallebuona_ernest.py U X p4_vallebuona_ernest.py U X JS tags.js X
assignments > cop4045 > Homework2 > p4_vallebuona_ernest.py > main
60 def display_top_actors(
97         else:
98             actor_money[actor] = money
99
100     ranking = sorted(
101         actor_money.items(),
102         key=lambda item: item[1],
103         reverse=True
104     )
105
106     print("\nTop Actors by Box Office\n")
107
108     rank = 1
109
110     for actor, money in ranking:
111         print(rank, actor, "$" + format(money, ",.0f"))
112         rank += 1
113
114     except Exception as fail:
115         print("Could not display top actors:", fail)
116         raise
117
118
119 def main() -> None:
120     """Test the functions."""
121
122     try:
123         casts_file = "imdb-top-casts.csv"
124         rated_file = "imdb-top-rated.csv"
125         grossing_file = "imdb-top-grossing.csv"
126
127         display_top_collaborations(
128             casts_file,
129             rated_file
130         )
131
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```
import csv
```

```
def display_top_collaborations(
```

```
    casts_filename: str, rated_filename: str
```

```
) -> None:
```

```
    """Display director-actor collaborations for top-rated movies."""
```

```
    try:
```

```
        rated_movies = set()
```

```
        with open(rated_filename, "r", encoding="utf-8") as file:
```

```
            reader = csv.DictReader(file)
```

```
            for row in reader:
```

```
                title = row["Title"]
```

```
                year = row["Year"]
```

```
                rated_movies.add((title, year))
```

```
        collaborations = {}
```

```
        with open(casts_filename, "r", encoding="utf-8") as file:
```

```
            reader = csv.reader(file)
```

```
            for row in reader:
```

```
title = row[0]
```

```
year = row[1]
```

```
director = row[2]
```

```
if (title, year) in rated_movies:
```

```
    actors = row[3:]
```

```
    for actor in actors:
```

```
        pair = (director, actor)
```

```
        if pair in collaborations:
```

```
            collaborations[pair] += 1
```

```
        else:
```

```
            collaborations[pair] = 1
```

```
ranking = sorted(
```

```
    collaborations.items(),
```

```
    key=lambda item: item[1],
```

```
    reverse=True
```

```
)
```

```
print("\nTop Director-Actor Collaborations\n")
```

```
rank = 1
```

```
for (director, actor), number in ranking:
```

```
print(rank, director, actor, number)
```

```
rank += 1
```

```
except Exception as fail:
```

```
print("Error displaying collaborations:", fail)
```

```
raise
```

```
def display_top_actors(
```

```
    casts_filename: str, grossing_filename: str
```

```
) -> None:
```

```
    """Display actors ranked by total box office from top-grossing movies."""
```

```
    try:
```

```
        grossing_movies = {}
```

```
        with open(grossing_filename, "r", encoding="utf-8") as file:
```

```
            reader = csv.DictReader(file)
```

```
            for row in reader:
```

```
                title = row["Title"]
```

```
                year = row["Year"]
```

```
                box_office = row["USA Box Office"]
```

```
                box_office = box_office.replace("$", "").replace(",", "")
```

```
grossing_movies[(title, year)] = float(box_office)
```

```
actor_money = {}
```

```
with open(casts_filename, "r", encoding="utf-8") as file:
```

```
    reader = csv.reader(file)
```

```
    for row in reader:
```

```
        title = row[0]
```

```
        year = row[1]
```

```
        if (title, year) in grossing_movies:
```

```
            money = grossing_movies[(title, year)]
```

```
            actors = row[3:]
```

```
            for actor in actors:
```

```
                if actor in actor_money:
```

```
                    actor_money[actor] += money
```

```
                else:
```

```
                    actor_money[actor] = money
```

```
ranking = sorted(
```

```
    actor_money.items(),
```

```
    key=lambda item: item[1],
```

```
    reverse=True
```

)

```
print("\nTop Actors by Box Office\n")
```

```
rank = 1
```

```
for actor, money in ranking:
```

```
    print(rank, actor, "$" + format(money, ",.0f"))
```

```
    rank += 1
```

```
except Exception as fail:
```

```
    print("Could not display top actors:", fail)
```

```
    raise
```

```
def main() -> None:
```

```
    """Test the functions."""
```

```
try:
```

```
    casts_file = "imdb-top-casts.csv"
```

```
    rated_file = "imdb-top-rated.csv"
```

```
    grossing_file = "imdb-top-grossing.csv"
```

```
display_top_collaborations(
```

```
    casts_file,
```

```
    rated_file
```

```
)
```

```
display_top_actors(
```

```
    casts_file,
```

```
    grossing_file
```

```
)
```

```
except Exception as fail:
```

```
    print("An error occurred during testing:", fail)
```

```
    raise
```

```
print("Ernest vallebyona: Z23588328 ")
```

```
if __name__ == "__main__":
```

```
    main()
```

Ran outta time, womp womp for me